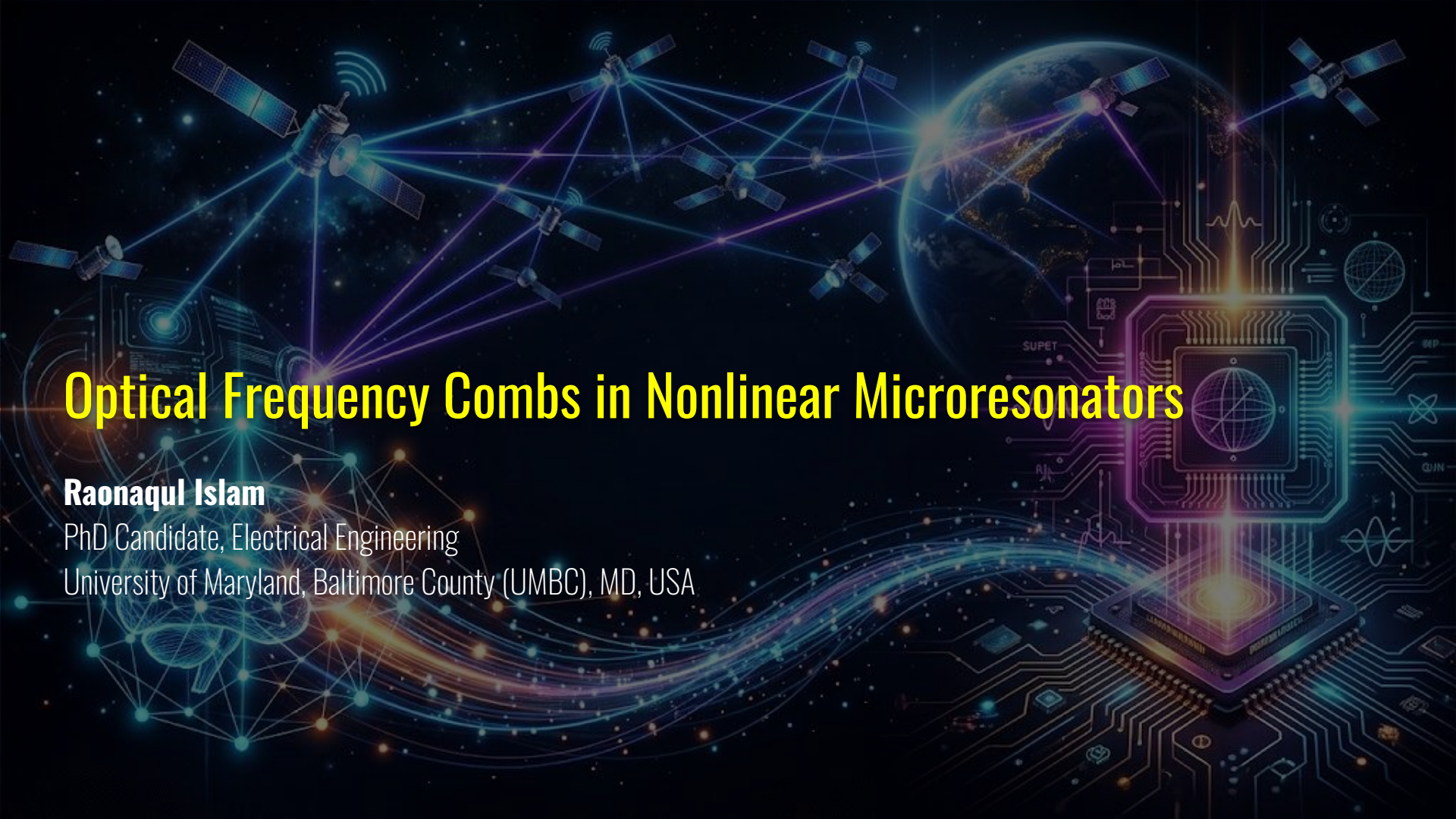


Optical Frequency Combs in Nonlinear Microresonators

Raonaqul Islam
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University of Maryland, Baltimore County (UMBC), MD, USA

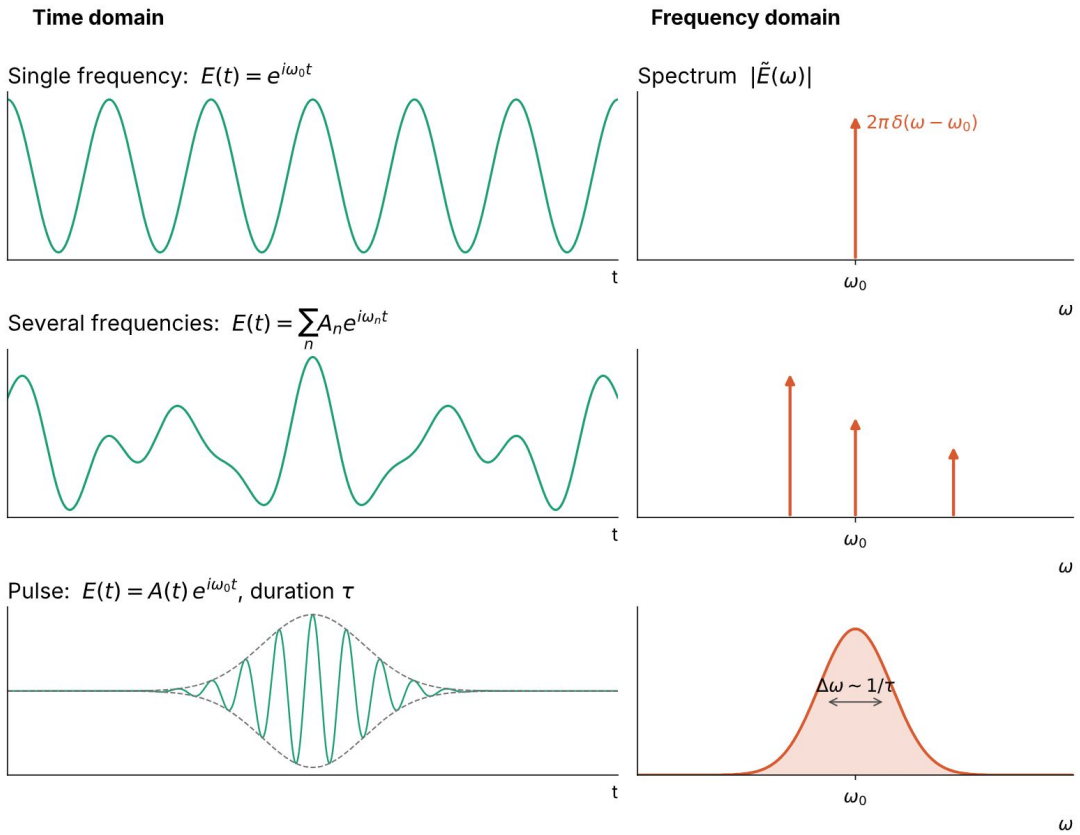
Outline

- Single frequency vs multiple frequencies (CW vs pulse)
- Optical frequency comb (OFC): definition and applications
- Concept review:
  - Dispersion
  - Nonlinearity – the Kerr effect
  - Gain and loss
- OFC in microresonator (microcomb) generation with dissipative Kerr soliton (DKS)
- Some physical properties of microcombs:
  - Laser scan vs self-starting
  - Importance of locking f-CEO and f-rep
  - Kerr induced synchronization (KIS)
  - Avoided mode crossings
- Generating a comb numerically using Lugiato Lefever Equation (LLE): a demonstration

Single frequency vs multiple frequencies

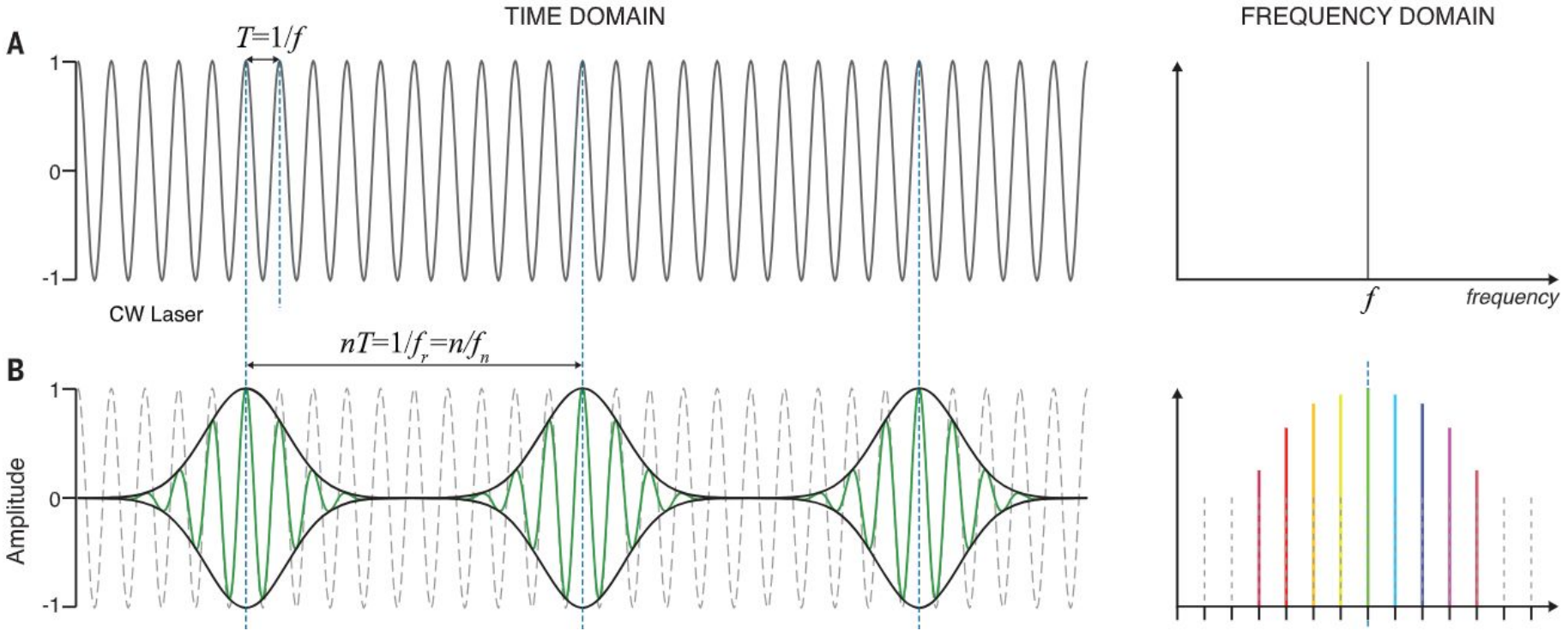
- Please take one minute and think about the figure on the right.
- After one minute, please divide into pairs and discuss it with your partner.
- We will have one volunteer from each group to say what they think.

01:00



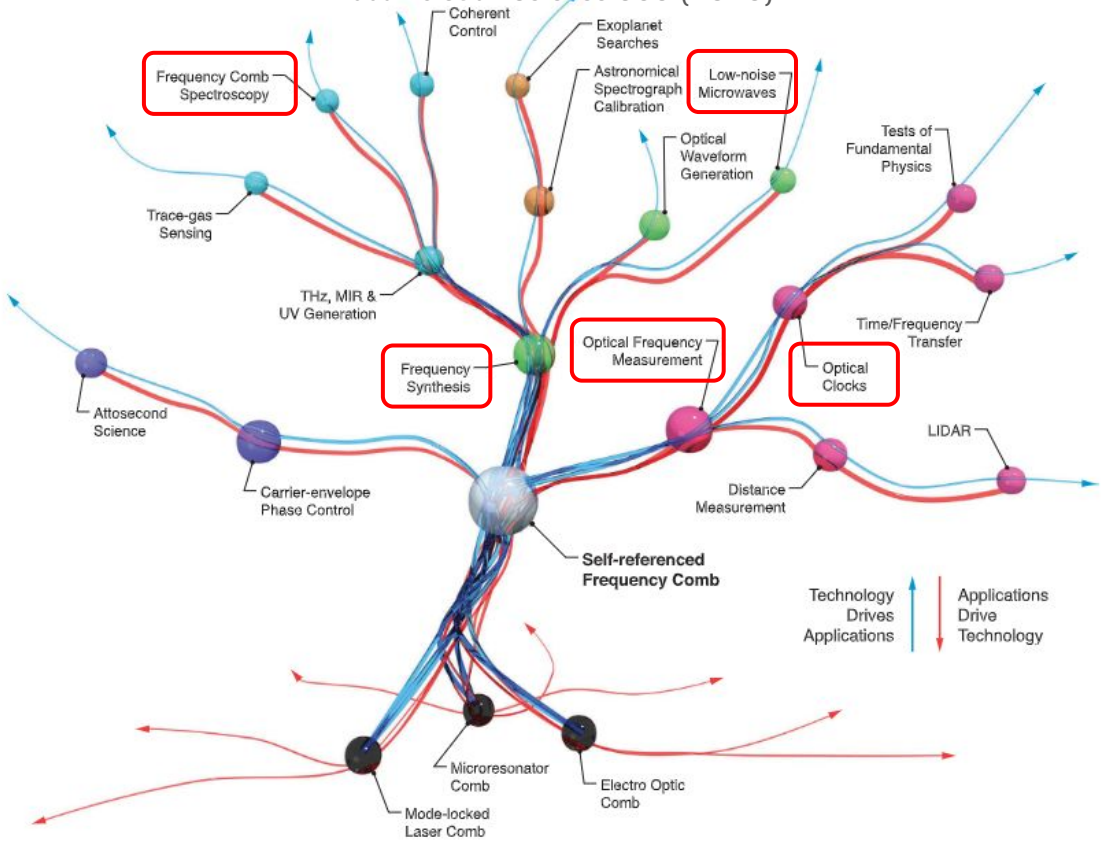
Optical Frequency Comb

Diddams *et al.* Science 369 (2020)



Optical Frequency Comb (OFC)

Diddams *et al.* Science 369 (2020)



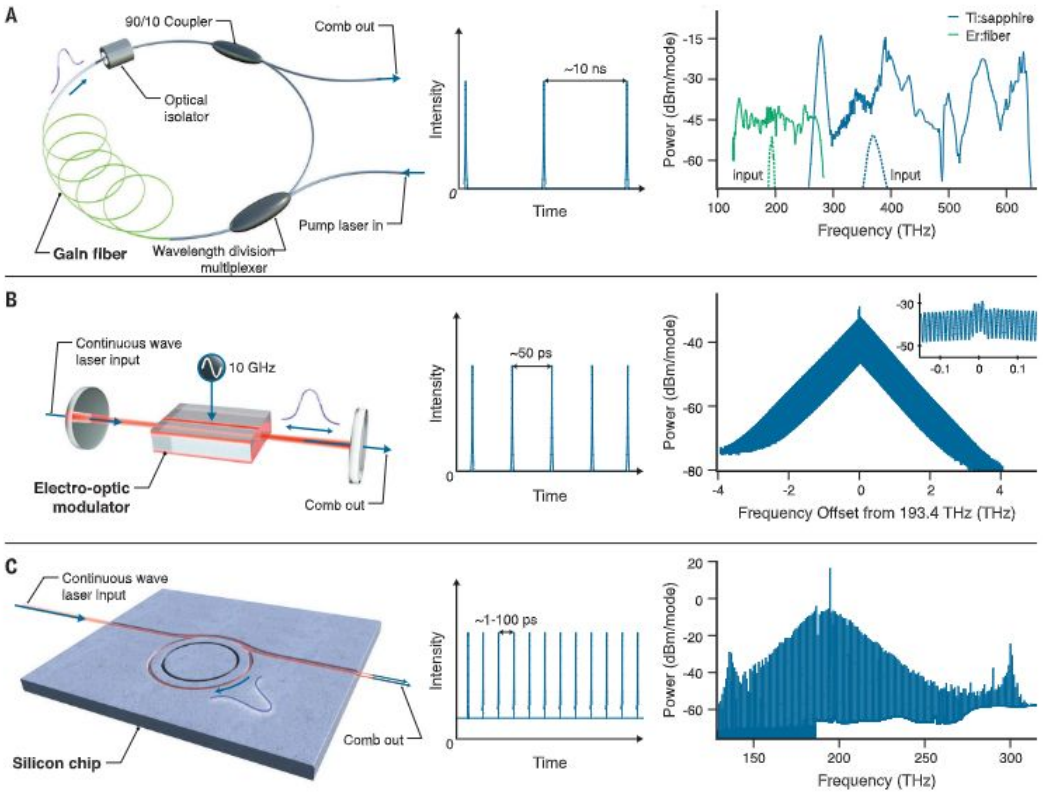
OFC Generation Methods

Mode-locked lasers

Electro-optic modulators

Microresonators

Diddams *et al.* Science 369 (2020)



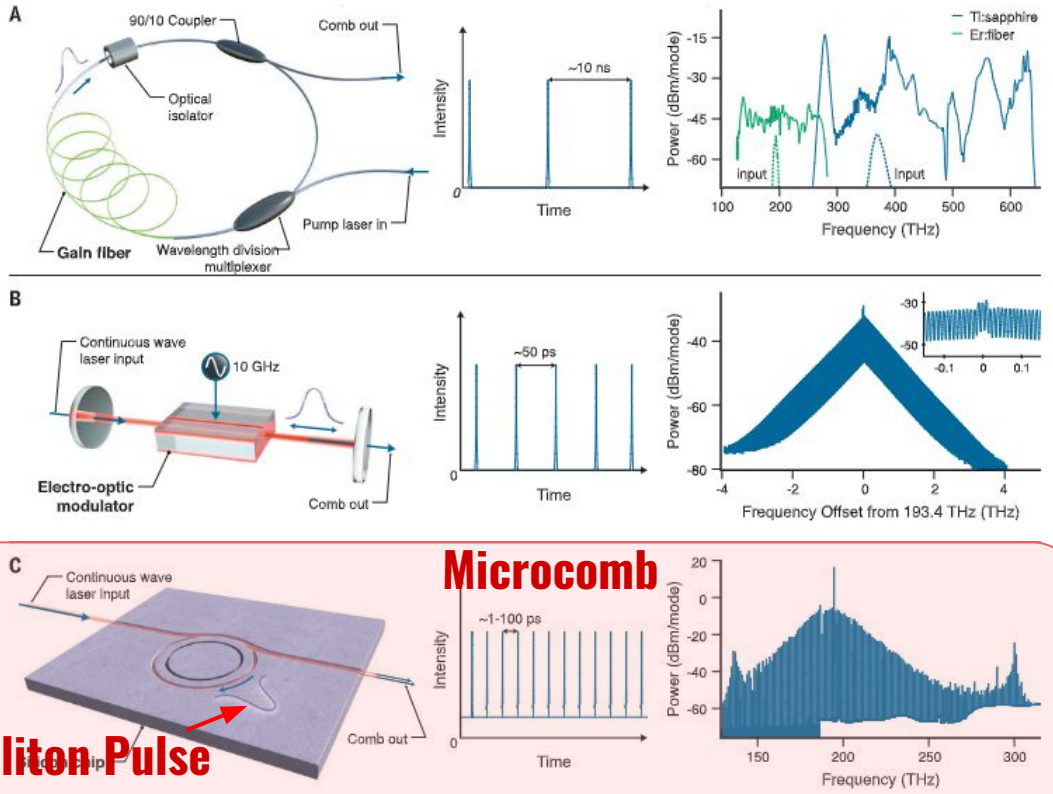
OFC Generation Methods

Mode-locked lasers

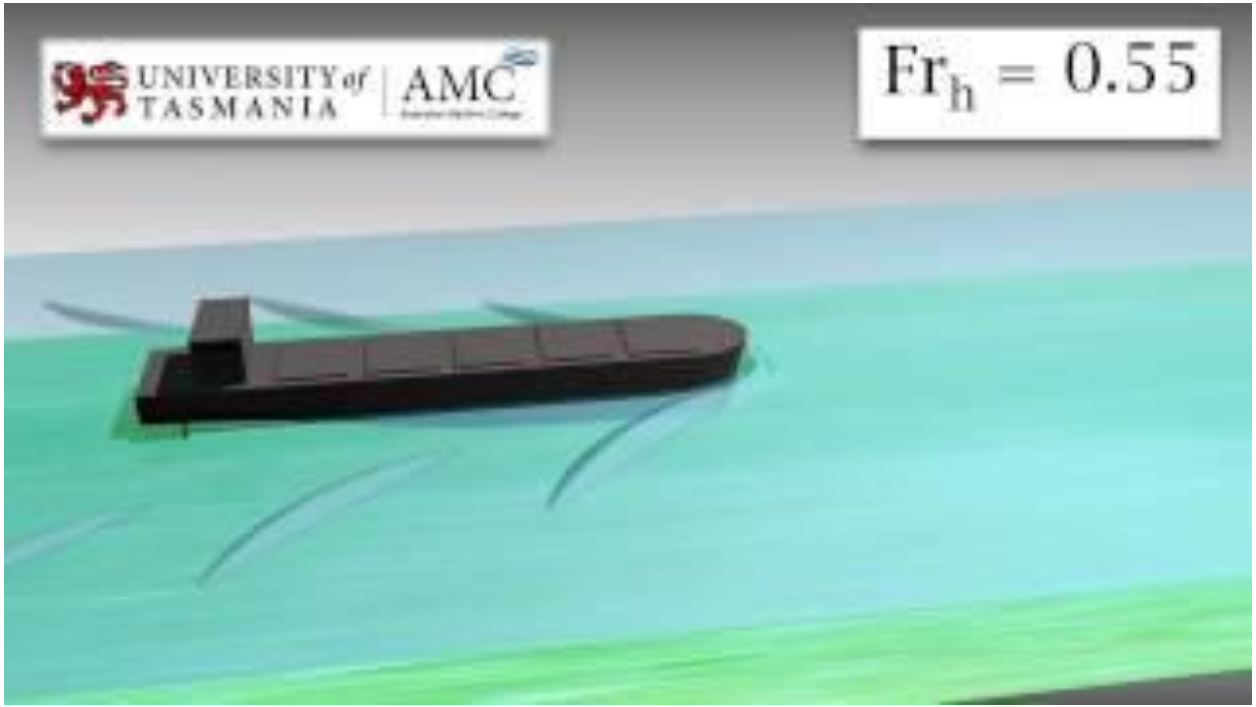
Electro-optic modulators

Microresonators

Diddams *et al.* Science 369 (2020)

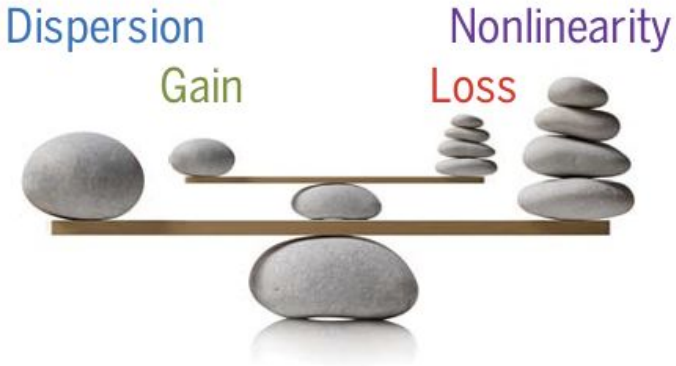


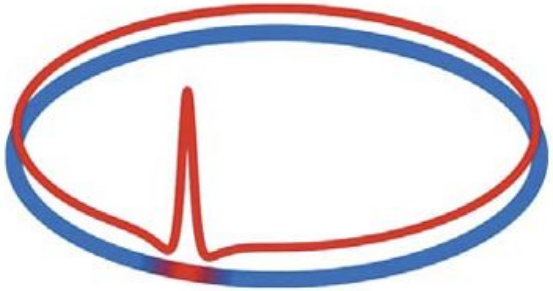
Solitary Waves

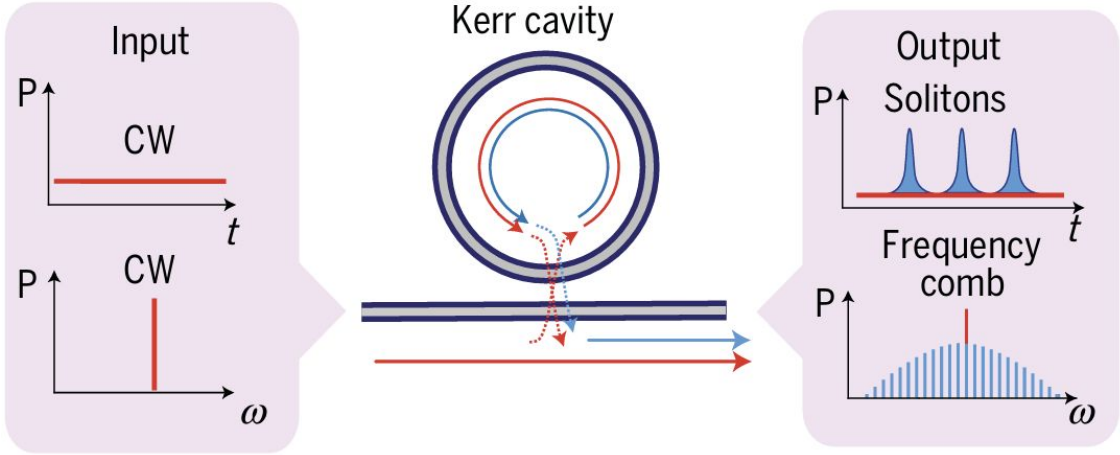


Soliton Pulse

Kippenberg *et al.* Science 361 (2018)

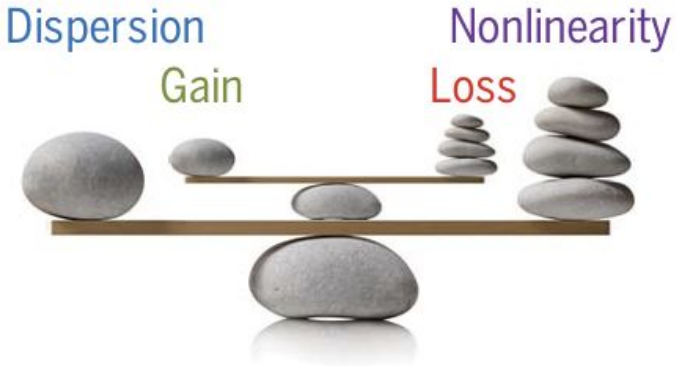


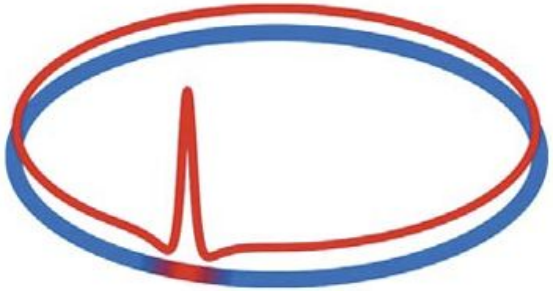


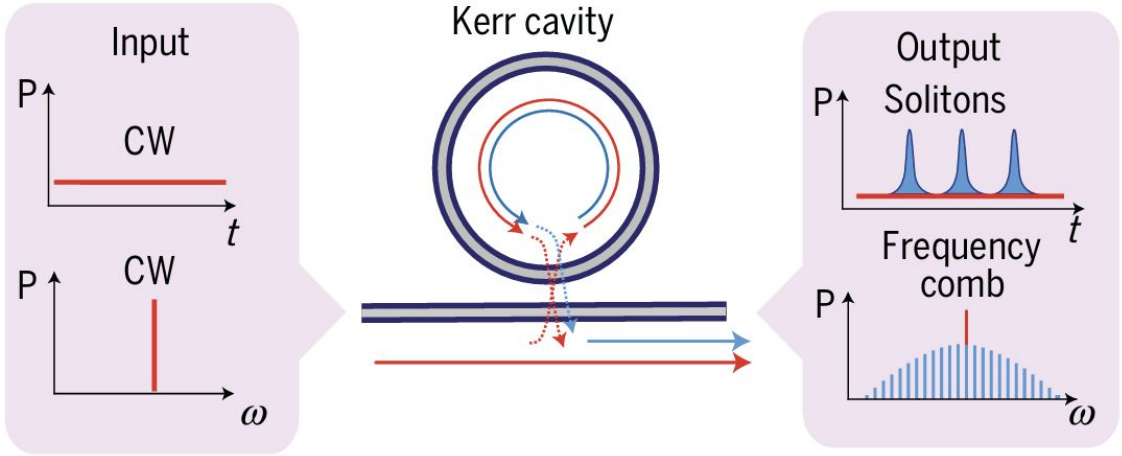


Soliton Pulse

Kippenberg *et al.* Science 361 (2018)







To understand how a soliton pulse is generated in the cavity, we need to review some concepts.

Concept Review: Dispersion

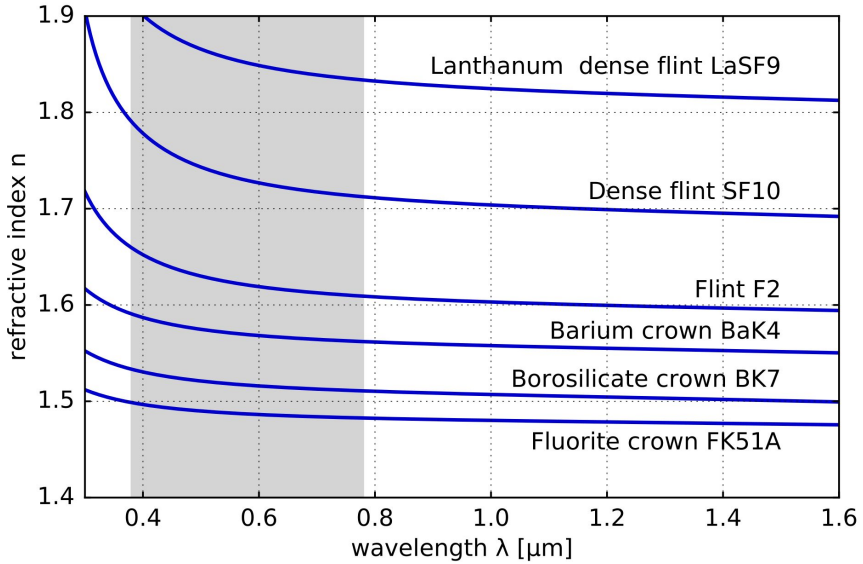
- Electromagnetic waves travel at speed of light
- In dielectric materials, this velocity is

  $v = c/n$

- Due to material and geometry, refractive index *n* changes with wavelength or frequency.

Conclusion: Different frequency waves travel at different speeds.

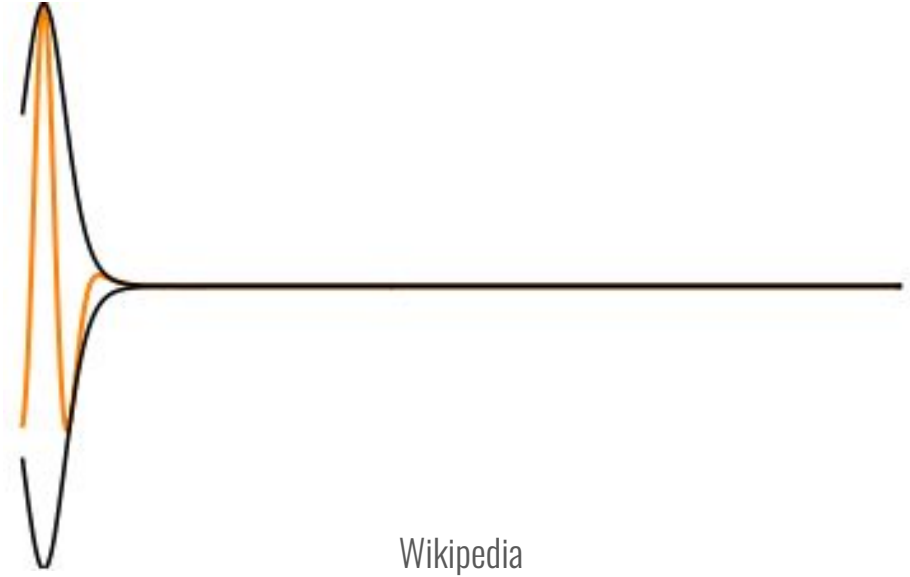
Wikipedia



Concept Review: Dispersion (contd.)

Discuss with your partner for two (02) minutes about what is happening in this movie.

02:00



Wikipedia

Concept Review: Kerr Nonlinearity

- Kerr nonlinear effect means – change in refractive index based on applied electric field.
- In optical or AC Kerr effect, electric field from the laser does the job.
- The changed refractive index becomes,

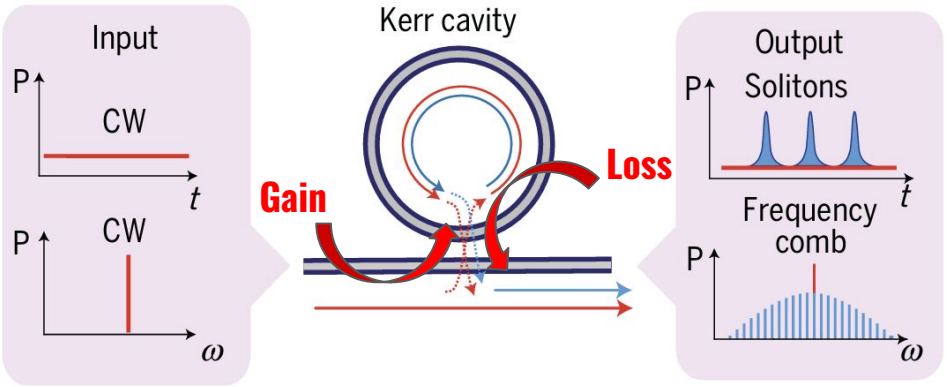
$$n = n_0 + n_2 I$$

- In a Kerr nonlinear medium, the pulse gets an intensity dependent phase due to nonlinearity.

Concept Review: Gain and Loss

- **Coupling loss:** some of the light fail to enter the microresonator.
- **Intrinsic loss:** happens internally in the microresonator.
- **Four-wave mixing:** light at three frequencies interact to produce light at a fourth frequency

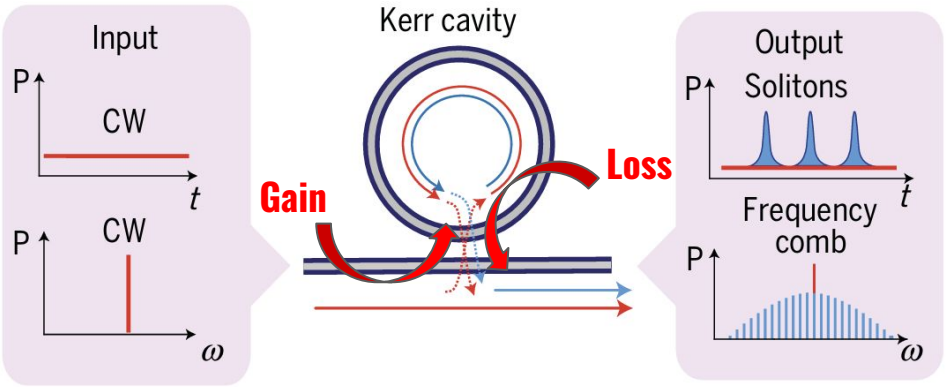
Kippenberg *et al.* Science 361 (2018)



Concept Review: Gain and Loss

- **Coupling loss:** some of the light fail to enter the microresonator.
- **Intrinsic loss:** happens internally in the microresonator.
- **Four-wave mixing:** light at three frequencies interact to produce light at a fourth frequency

Kippenberg *et al.* Science 361 (2018)



Loss can be balanced by gain through four-wave mixing!

Concept Summary: Two-Minutes Study

Dispersion
- Different frequencies have different velocities
- Changes pulse shape

Kerr Nonlinearity
- Refractive index changes with intensity
- Gives the pulse intensity dependent phase

Loss
- Coupling loss of power
- Loss intrinsic to the microresonator

Gain
- Comes from the laser pump
- Can compensate loss via four-wave mixing

02:00

Short Quiz

A pulse is propagating in a Silica waveguide. The pulse changed its shape along the way of travelling. Which effect from the list caused this shape change?

a) Kerr nonlinearity
b) Gain
c) Loss
d) Dispersion

Short Quiz

A pulse is propagating in a Silica waveguide. The pulse changed its shape along the way of travelling. Which effect from the list caused this shape change?

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The Ultimate Balance



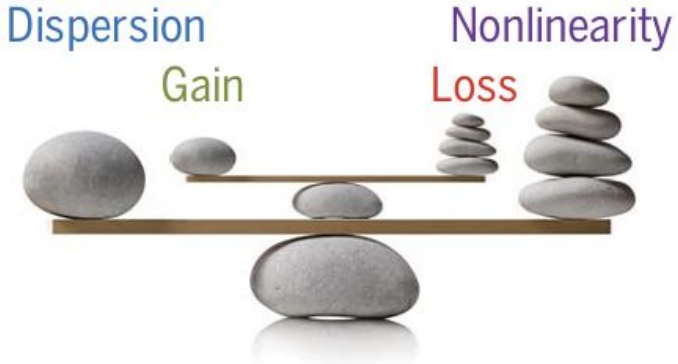
Google

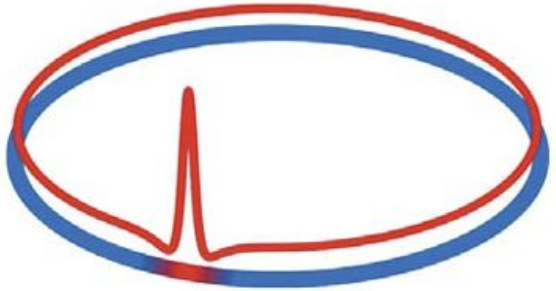
The Ultimate Balance



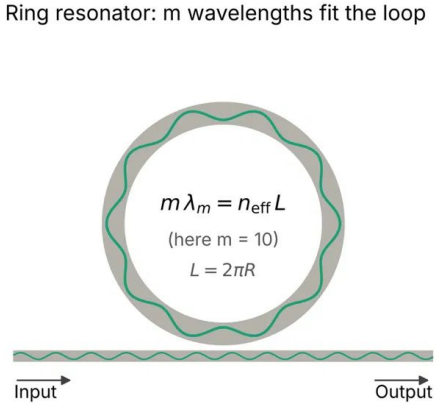
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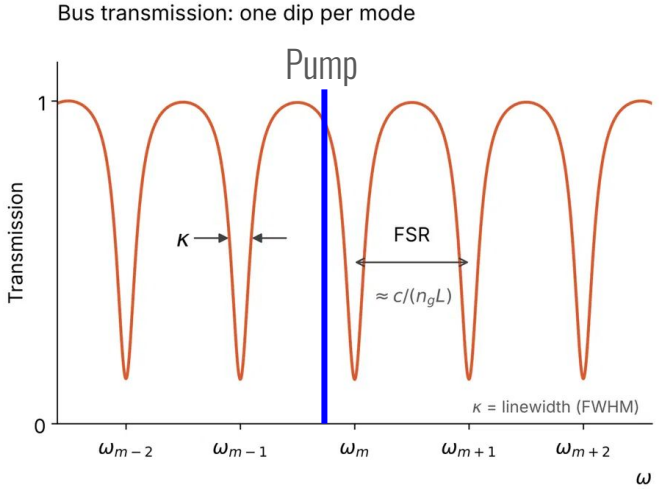
Kippenberg *et al.* Science 361 (2018)

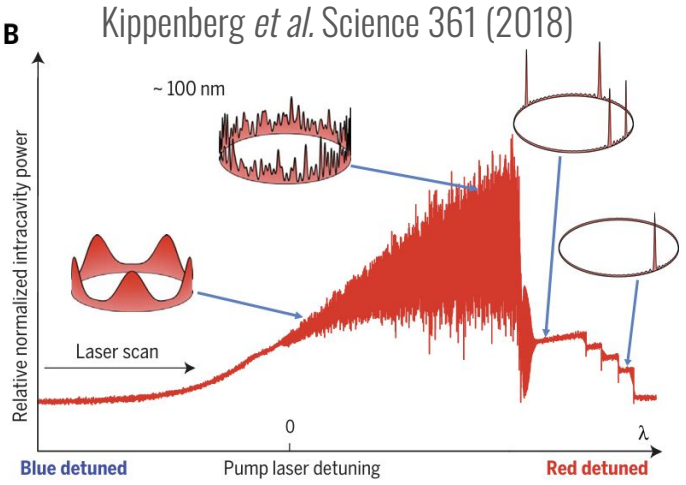




Laser Scan Method



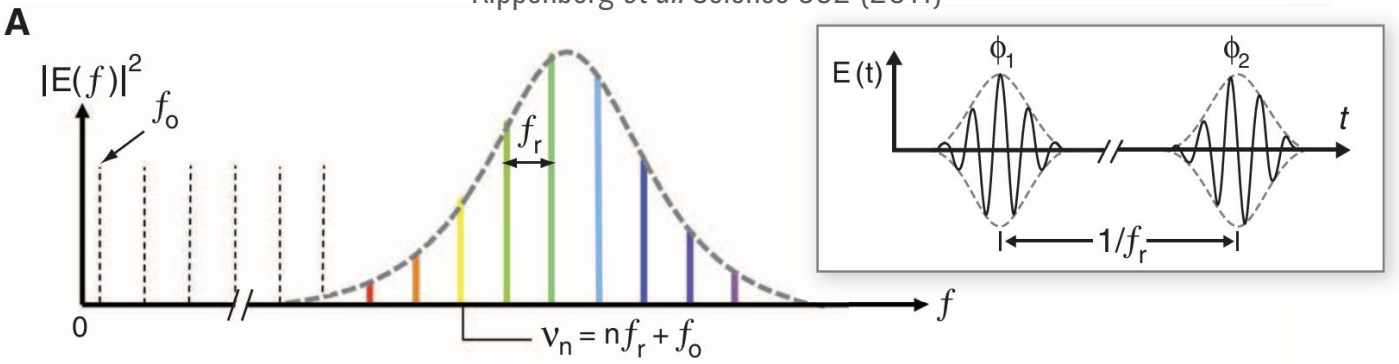




- Microresonators have resonant frequencies – geometric and material property. Gap between resonances – FSR.
- At **Laser Frequency = Resonant Frequency**, maximum power is coupled.
- To generate solitons, laser power is slightly detuned from the nearest resonant frequency.
- It is difficult to determine the exact required detuning, so the laser is scanned from blue to red detuned region until a soliton is obtained.
- **Generating solitons without laser scanning is an interesting research area.**

Importance of f_{CEO} and f_{rep}

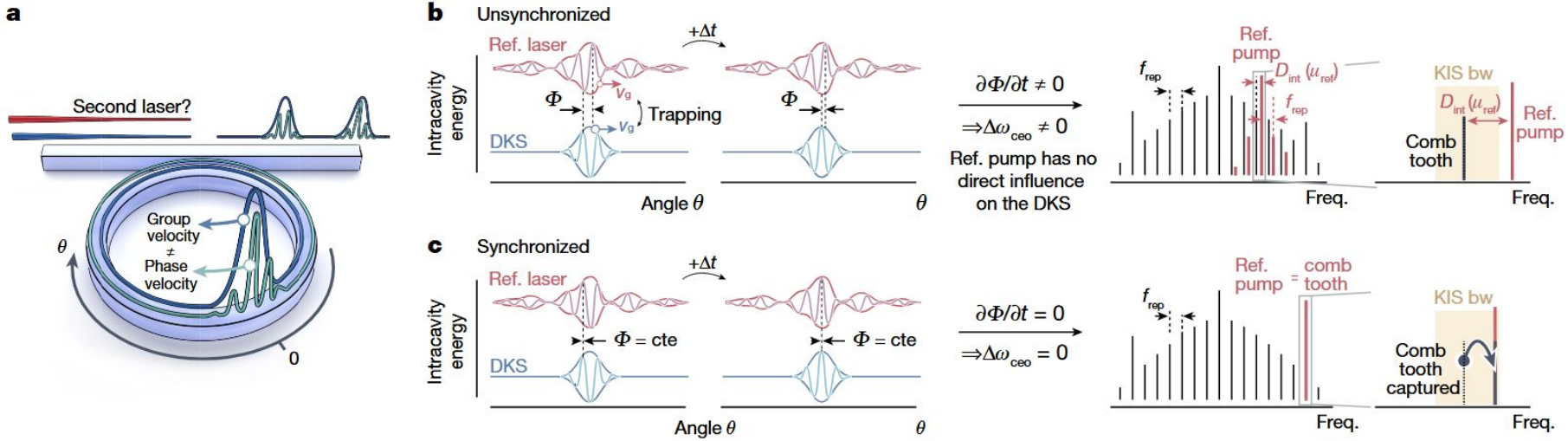
Kippenberg *et al.* Science 332 (2011)



- f_{CEO}: Carrier envelope offset (f_0)
- f_{rep}: repetition-rate (f_r)
- Fluctuations in f_{CEO} complicates synthesizer applications
- Fluctuations in f_{rep} complicates microwave generation
- Fascinating research is being performed to lock both of these quantities

Kerr Induced Synchronization (KIS)

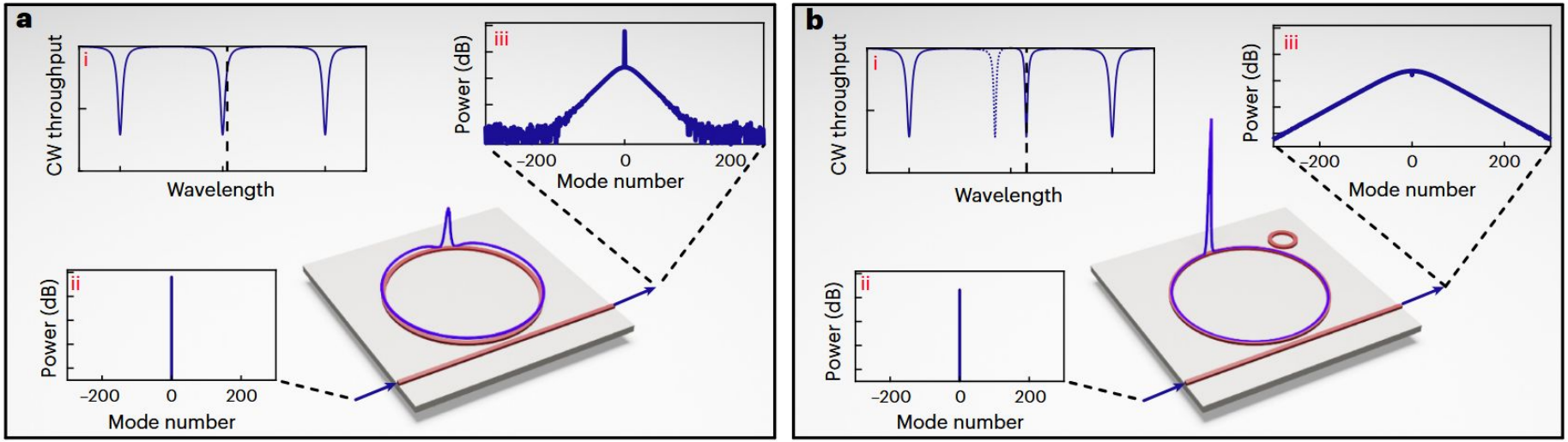
Moille *et al.* Nature 624 (2023)



- A second laser is sent – phase locks with one comb line.
- A fascinating discovery to lock f_{CEO} and f_{rep}.

Avoided Mode Crossings

Helgason *et al.* Nat. Photon. 17 (2023)



- The resonator resonances are modified by either defects or another ring – avoided mode crossing (AMX).
- Useful to improve power conversion efficiency in microcombs.

Soliton Modeling: Normalized Lugiato-Lefever Equation (LLE)

Field — Detuning — Nonlinearity — Dispersion — Pump

$$\frac{\partial \psi}{\partial \tau} = -(1 + i\alpha)\psi + i\gamma |\psi|^2 \psi - i\frac{\beta_2}{2}\frac{\partial^2 \psi}{\partial \theta^2} + F$$

- The above normalized equation is solved numerically.
- The solution, ψ, represents the dissipative Kerr Soliton.
- Fourier transform of a train of soliton pulses gives the frequency comb.

